# FLORENT TARIOLLE

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*Seeking a six-month research internship abroad in world models, starting April 2027.*

## Education

**INSA Rouen Normandy** 2023 – 2027
*Engineering degree, Computer Science and Artificial Intelligence; expected 2027*

**University of Rouen Normandy** 2026 – 2027
*Master's degree, Machine Learning for Artificial Intelligence; expected 2027*

**Lycée Dupuy de Lôme** 2021 – 2023
*Two-year intensive preparatory programme in mathematics and physics (CPGE MP)*

## Publications

**DashVMC: Real-Time Discrete World Model Control in Geometry Dash** Sep 2026
*F. Tariolle and F. Yger. PTA workshop, NeurIPS 2026 — accepted paper; poster.*

**arXiv: 2609.40003** · **Demo** · **Code**

- Developed a world-model agent for live Geometry Dash control from pixels, refining a behaviour-cloned policy with PPO entirely in frozen-model rollouts.
- Trained offline on approximately two hours of gameplay; improved survival over cloning on three official levels and a held-out layout, with 60-FPS control on an RTX 2060.

**Opportunistic Target Selection: Early Directional Commitment for Query-Efficient Black-Box Adversarial Attacks** May 2026
*F. Tariolle and F. Yger. CAp 2026 (Conférence sur l'Apprentissage automatique) — accepted paper; poster.*

**arXiv: 2605.25663** · **Code** · **Poster**

- Developed a wrapper for score-based black-box adversarial attacks that uses early class-confidence signals to select a target, reducing class drift without gradient access.
- Evaluated 4,500 ImageNet attack runs across five classifiers. On ResNet-50, improved attack success by up to 27 percentage points and reduced censored-mean attack iterations by 43%.

## Research Projects and Awards

**Finalist, Hack the World(s) Hackathon** Jun 2026
*Geometry-aware joint-embedding EEG · 24-hour JEPA and world-model challenge*

**Code** · **Presentation**

Selected as one of 100 participants from 650+ applicants for a hackathon sponsored by Yann LeCun. Our team finished among the top five of 25 teams.

Built a model that learns from unlabeled EEG recordings. Explored whether training that respects the curved geometry of covariance matrices leads to more structured representations.

## Experience

### InterDigital

**Generative AI Consultant** Oct 2026 – Present

*Freelance · France, remote*

Advising InterDigital's R&D teams on generative AI workflows, plugins, and skills, building on tools I initiated during my research internship. Helping teams develop and maintain these tools independently through practical guidance on AI-assisted development, reliability, and security.

**Deep Learning Research Intern** May – Sep 2026

*Rennes, France*

Researched learned decoder-side filter selection as an alternative to neural in-loop filtering for VVC (H.266) video decoding, with the aim of reducing compression and quantization artifacts. Contributed to more than 10 patent applications arising from this research.

### Enedis

**Machine Learning Intern** Jan 2026 – Present

*Part-time capstone project, 25 hours/week · Rouen, France*

Building a digital twin of Enedis's distribution network by mapping synthetic customer load profiles to distribution substations and linking these to primary substations. Combining DeepCourboGen-generated curves with measured substation loads, and developing spatial and temporal adjustments for more realistic local profiles.

*Paused May–September 2026 during the research internship at InterDigital.*

## Open-Source Projects

**VisualTorch** Jul 2026 – Present

*Maintainer and MCP Integration Lead · Official PyTorch Ecosystem project*

Maintaining a library for visualizing PyTorch neural-network architectures. Designed and implemented its MCP integration to generate editable diagrams from model definitions, making architecture inspection available through agent tooling. The project has more than 300 GitHub stars.

**Rose — Real-time customization for League of Legends** Sep – Dec 2025

*Co-founder and Lead Developer · Open source*

Co-founded and co-developed an open-source customization tool; remain the project's top contributor. Built core application components spanning the Python backend, in-client JavaScript plugins, and their real-time WebSocket bridge.

The project has since reached 30,000 daily active users and 90,000 monthly active users, with 700+ GitHub stars and 20+ contributors.

## Technical Skills

**Programming:** Python, C++
**Frameworks:** PyTorch
**Research:** Self-supervised learning, world models, model-based reinforcement learning and planning
**Languages:** French (native), English (TOEIC 970/990), Spanish (intermediate)

## Interests

League of Legends (top 0.1% of ranked players; two LAN wins), bouldering, and strength training.

## References

**Florian Yger** (INSA Rouen Normandy)

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**Frédéric Lefèbvre** (InterDigital)

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